Chapter V – A Review of Vectors

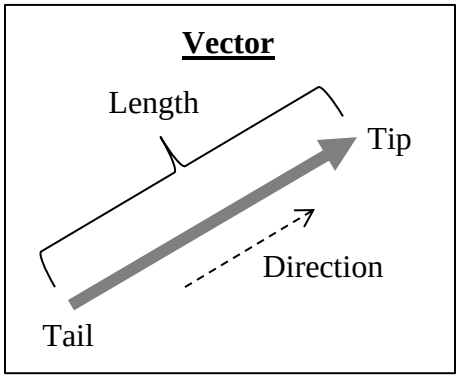
Calculus Physics II

Scalar - is a physical quantity that can be entirely described with a single number (including units). To understand the meaning of a scalar quantity all that is needed is the amount and the unit. We deal with scalars using traditional algebra.

Examples: – time (t = 5 s), mass (m = 4.2 kg), volume (V = 1.3 m^3), temperature (T = 27 °C), Odometer on car (speed = 35 mph), etc. …

Vectors in general vectors can be thought of as an array of numbers or a set of elements that are required to describe physical quantities. For a vector, it is not only the amount of the variable that matters but also the direction or orientation of the vector. A vector is a mathematical representation of a physical quantities. We deal with vectors in mathematics using vector algebra and vector calculus which is different from their scalar versions.

- The vector is characterized by the length of the arrow which indicates the amount of the measured quantity as a number (and unit) and the arrow indicates its direction of the vector.
- Vectors are objects that reside in three dimensions therefore they are generally represented by three elements (or components).



The **Length** of vector is proportional to the amount of the vector and is given in the units of the quantity being measured.

The **Arrow** points in the direction the vector is acting.

The base of the vector is called the "Tail" and the end with the arrow is called the "Tip".

- Vectors are equal when both their length and direction are the same and they have the same units.
- If they have different lengths or they have a different direct the vectors are not the same vector, they are not equivalent!

Indicating a vector quantity (arrow vector) - A vector quantity is indicated by an arrow above the symbol or the letter being "bolded", it is **VERY IMPORTANT** to properly mark vectors because mathematically vectors are treated differently than scalars. Examples are:

The Force vector is $\vec{\mathbf{F}}$ The velocity vector is $\vec{\mathbf{v}}$

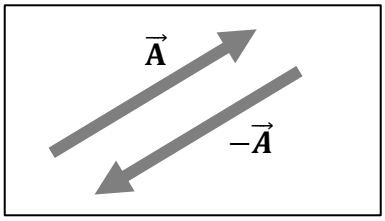
Magnitude (length, amount or norm) of a vector – is the length of the vector in vector space. It is the "how much" of the vector or the amount or value of the vector, with no direction associated with it is a scalar quantity. It is also known as the norm of the vector. The magnitude of the vector is always a positive value. The term Magnitude applies to any type of vector!

The magnitude of the vector is indicated by the absolute value signs around the vector and is the scalar quantity of the vector.

The magnitude of vector $\vec{\mathbf{a}}$ is given by $|\vec{a}|$.

- Magnitude is ALWAYS a positive number.

The negative of a vector – Multiplying a vector by negative one (-1) flips the direction of the vector 180° so it points in the opposite direction but does not change the length or magnitude of the vector. The new vector is a different vector because it has a different direction.



Unit vector – The unit vector is a vector that represents the direction of the vector only! It is the normalized length of the full vector.

- The unit vector is indicated by a "carrot" or "hat" placed over the vector symbol $\widehat{\mathbf{A}}$ **.**
- The unit vector has no units.
- The unit vector of an arrow vector is determined by dividing the vector by its length.
- The unit vector has a magnitude that is equal to 1 and points in the same direction as the vector and opposite direction if the scalar position of the vector is a negative.

$$\left|\hat{A}\right| = 1 \ \ for\ all\ unit\ vectors$$

The unit vector $\widehat{\mathbf{A}}$ of a given vector $\vec{\mathbf{A}}$ can be found by dividing the vector itself by its magnitude, this gives the unit vector with a magnitude of one and indicates the direction of the vector.

$$\text{The unit vector } \widehat{\mathbf{A}} = \frac{\vec{\mathbf{A}}}{\left|\vec{\mathbf{A}}\right|}$$

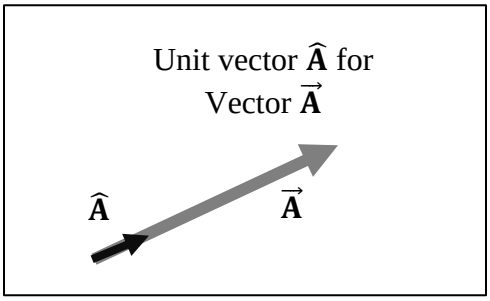
- The unit vector $\widehat{\mathbf{A}}$ is one scaled "unit" of the vector $\vec{\mathbf{A}}$ in the direction of vector $\vec{\mathbf{A}}$**.**

A vector is written as a magnitude (amount of the vector or length) and direction (the unit vector)

$$\vec{\mathbf{A}} = (magnitude)\widehat{\mathbf{Direction}}$$

or

$$\boxed{\vec{\mathbf{A}} = (|\vec{\mathbf{A}}|)\hat{\mathbf{A}}}$$



- ❖ The unit vector gives the "scale" of the vector's length along with its direction.

  - The unit vector is a vector.
  - The unit vector has a length or magnitude equal to 1.

Scalar components of a vector have magnitude only and do not include direction, although the sign of the component indicates whether it points in the positive or negative direction of the chosen axis and same or opposite direction than the unit vector. Breaking vectors into scalar components is an essential technique in physics because it simplifies the analysis of motion, forces, and other vector quantities by allowing each direction to be considered independently.

Mathematical operations on vectors – Vectors are a mathematical representation of a physical quantity therefore vectors can be added, subtracted and multiplied but there is specific rule for mathematical operations on vectors (adding and multiplying vectors).

- ➢ One can NOT divide by a vector!

Summation of vectors (Σ): The summation of vectors is the vector addition of all vectors being added.

$$\sum_{i=1}^{n} \vec{\mathbf{A}}_{\mathbf{i}} = \vec{\mathbf{A}}_1 + \vec{\mathbf{A}}_2 + \vec{\mathbf{A}}_3 + \cdots \vec{\mathbf{A}}_n$$

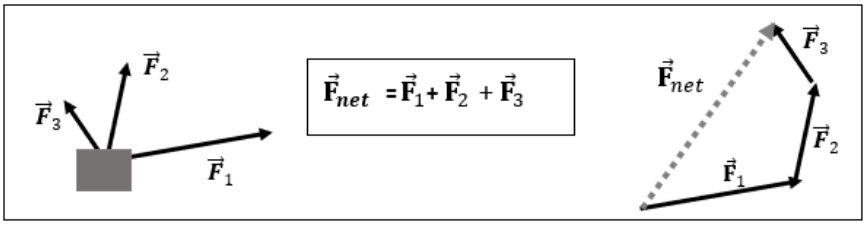
The Net Force acting on an object is the vector sum of all forces acting on that object.

$$\mathbf{Net\ force} = \Sigma\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 + \vec{\mathbf{F}}_3 + \cdots \vec{\mathbf{F}}_n = \vec{\mathbf{R}} = \vec{\mathbf{F}}_{net}$$

The image below shows three forces acting on a box, the net force acting on the box is the vector sum of all forces acting on the box. These forces can be arranged to add them graphically and find the net force from the vector addition of the forces.

The magnitude and direction of the net force is the magnitude and direction of a single force ($\vec{\mathbf{F}}_{net}$) that would affect the box in exactly the same manner as the three forces combined.

- The sum of the vectors is not the sum of the magnitudes of the vector because direction of the vectors matter!



Change in vectors (Δ): The change in vector quantities is given by the final vector subtracted from the initial vector or the vector sum of the final vector plus the negative of the initial vector.

$$\Delta\vec{\mathbf{A}} = \vec{A}_f - \vec{A}_i$$

$$\Delta\vec{\mathbf{A}} = \vec{A}_f + (-\vec{A}_i)$$

Multiplying Vectors –There are very specific rulers for multiplying vectors using the different methods of vector multiplication. Multiplication of vectors is also called a vector Operation. We have already seen an operation on vectors, this would be adding vectors, addition can be thought of as an operator.

The multiplication operators have different results depending on which operator is used. The type of multiplication that is used depends on the specific quantities in Physics and their properties. There are 3 types of multiplication operations (or operators) that can be performed on vectors:

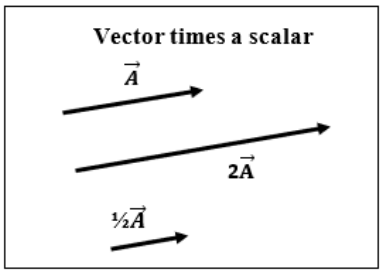
1) Multiply/divide a vector by a scalar that results in a new vector.
2) Dot Product results in a scalar quantity
3) Cross Product results in a new vector

The method of multiplication depends on why they are being multiplied and the end result of the multiplication.

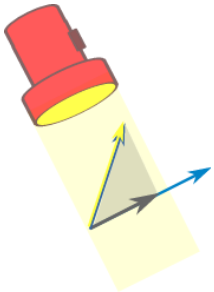
Multiplying a vector by a Scalar (constant) – when a vector is multiplied by a constant then the constant is multiplied by each of the component vectors and is still a vector. Assume "s" is a constant then vector $\vec{\mathbf{A}}$ times constant "s" is

$$s\vec{\mathbf{A}} = \left(s\left|\vec{\mathbf{A}}\right|\right)\hat{\mathbf{A}}$$

$$s\vec{\mathbf{A}} = (sA_x)\hat{\boldsymbol{x}} + (sA_y)\hat{\boldsymbol{y}}$$



Multiplying Vectors using the Dot Product ("•") – The multiplication of vectors using the Dot product produces a scalar quantity, no direction associated with the value. The Scalar product is the magnitude of the projection of the first vector $\vec{A}$ on to the second vector multiplied by the length of the second vector $\vec{B}$.



- The Dot Product of vector $\vec{A}$ onto vector $\vec{B}$ is the amount of vector $\vec{A}$ that is along the direction of vector $\vec{B}$ multiplied by the length of vector $\vec{B}$.
- The Dot Product indicates the amount of interaction that vector $\vec{A}$ has with vector $\vec{B}$ or vice-versa.

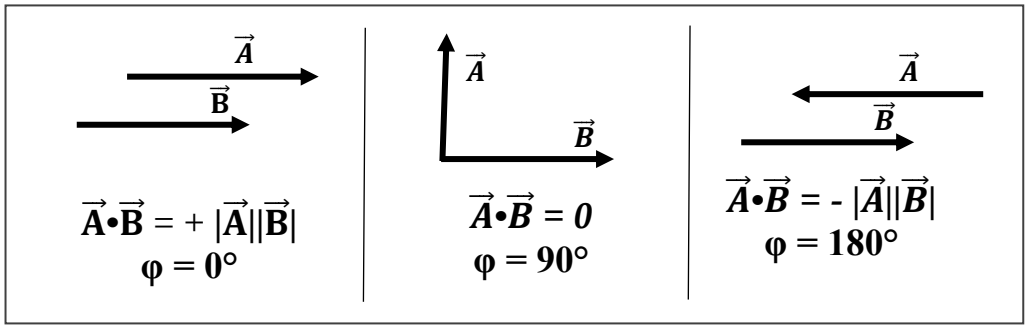
The Dot Product produces a scalar quantity, it is a number with a unit, there is no direction associate with the Dot Product!

$$\vec{A}\bullet\vec{B} = |\vec{A}||\vec{B}|\cos\varphi$$

$|\vec{A}|$ - is the magnitude of vector $\vec{A}$
$|\vec{B}|$ - is the magnitude of vector $\vec{B}$
Angle phi (φ) – is the **angle between the two vectors.**

The Dot Product of vector $\vec{A}$ on to vector $\vec{B}$ can be thought of as "How much does vector $\vec{A}$ affect vector $\vec{B}$?" "What influence does vector $\vec{A}$ along the line of vector $\vec{B}$?"

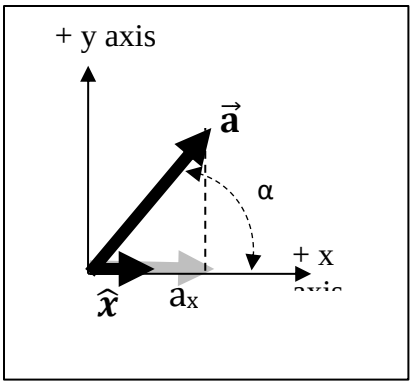
Angle between the vectors phi (φ) – The angle phi is the angle the two vectors in the dot product make with each other.



Components of a Vector – Components of a vector are vectors themselves that are parts of a larger vector, the components of the vector are the scalar projection of that vector along the principle axis and are made up of a scale component multiplied by the unit vector. When the components are added (vector addition) they create the complete vector. The scalar components of the vector are the amount or projection of the vector along a principle axis.

Given the magnitude and direction of vector $\vec{a}$, the vector can also be represented in vector component form as

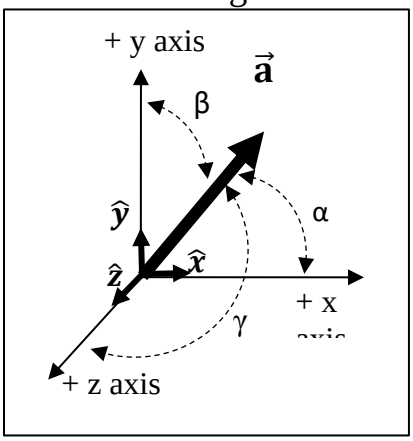
$$\vec{\mathbf{a}} = (a_x)\hat{\mathbf{x}} + (a_y)\hat{\mathbf{y}} + (a_z)\hat{\mathbf{z}} = (|\vec{\mathbf{a}}|)\hat{\boldsymbol{a}}$$



where $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ are the unit vectors along their respective axes and a_x, a_y and a_z are the scalar components of the vector along these axis' and are mutually perpendicular to each other.

- ✓ A component consists of a scalar multiplied by a <u>unit vector</u> along one of the principle axis. This component then points along a specific axis with a magnitude determined by the scalar.

Given the vector below in the Cartesian Coordinate system and the unit vectors along the positive principle axis.

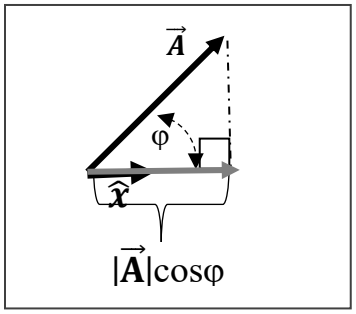


$(a_x)\hat{\mathbf{x}}$ is a vector with a direction along the x-axis ($\hat{\mathbf{x}}$) and a scalar value of a_x.
$(a_y)\hat{\mathbf{y}}$ is a vector with a direction along the y-axis ($\hat{\mathbf{y}}$)and a scalar value of a_y.
$(a_z)\hat{\mathbf{z}}$ is a vector with a direction along the z-axis ($\hat{\mathbf{z}}$)and a scalar value of a_z.

- Each unit vector makes an independent vector along a principle axis and the addition of all unit vectors (with scalar components) makes up that complete vector.
- A negative scalar component means that the direction is along the negative axis direction.

<u>Determining Components of vectors using the Dot Product and the unit vectors</u> – In order to find the components (projection along a principle axis) the Dot Product is used. By taking the Dot Product of the vector with the unit vector along a principle axis, the projection of the vector along that principle axis (or that unit vector) is produced. This technique can be very useful for determining properties of vectors.

$$\vec{\mathbf{A}} \bullet \vec{\mathbf{x}} = |\vec{\mathbf{A}}||\hat{\boldsymbol{x}}|\cos\varphi = |\vec{\mathbf{A}}|\cos\varphi$$



$|\vec{\mathbf{A}}|$ - is the magnitude of vector $\vec{\mathbf{A}}$
$|\hat{\boldsymbol{x}}|$ - is the magnitude of unit vector $\hat{\boldsymbol{x}}$ and is equal to 1.
Angle phi (φ) – is the **<u>angle between the two vectors.</u>**

<u>Projection of a vector along the principle axis.</u> Determine the projection of vector $\vec{\mathbf{a}}$ along the principle axis using the dot product.

Taking the Dot product of vector $\vec{\mathbf{a}}$ with the unit vector $\hat{\mathbf{x}}$

$\vec{\mathbf{a}} \bullet \hat{\boldsymbol{x}} = |\vec{\mathbf{A}}||\hat{\boldsymbol{x}}|\cos\varphi$ Using the fact that $|\hat{\boldsymbol{x}}| = \mathbf{1}$

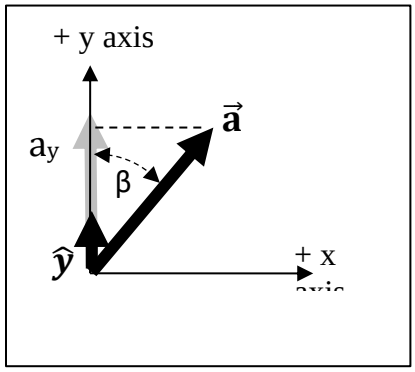
$$\vec{\mathbf{a}} \bullet \hat{\boldsymbol{x}} = a_x = |\vec{\mathbf{a}}|\boldsymbol{cos}(\boldsymbol{\alpha}) \quad \textbf{x scalar component of vector } \vec{\mathbf{a}}$$

- Where the angle alpha (α) is measured from the unit vector $\hat{\mathbf{x}}$ to the vector $\vec{\mathbf{a}}$. This is measured from the positive x-axis to the vector $\vec{\mathbf{a}}$.

Taking the Dot product of vector $\vec{\mathbf{a}}$ with the unit vector $\hat{\mathbf{y}}$

$\vec{\mathbf{a}} \bullet \hat{\boldsymbol{y}} = |\vec{\mathbf{A}}||\hat{\boldsymbol{y}}|\cos\varphi$ Using the fact that $|\hat{\boldsymbol{y}}| = \mathbf{1}$

$$\vec{\mathbf{a}} \bullet \hat{\boldsymbol{y}} = a_y = |\vec{\mathbf{a}}|\boldsymbol{cos}(\boldsymbol{\beta}) \quad \textbf{y scalar component of vector } \vec{\mathbf{a}}$$



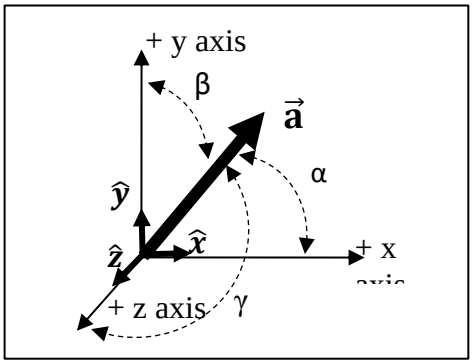
- Where the angle beta (β) is measured from the unit vector $\hat{\mathbf{y}}$ to the vector $\vec{\mathbf{a}}$. This is measured from the positive y-axis to the vector $\vec{\mathbf{a}}$.

Taking the Dot product of vector $\vec{\mathbf{a}}$ with the unit vector $\hat{\mathbf{z}}$ $\quad \vec{\mathbf{a}} = (a_x)\hat{\mathbf{x}} + (a_y)\hat{\mathbf{y}} + (a_z)\hat{\mathbf{z}}$

$\vec{\mathbf{a}} \bullet \hat{\boldsymbol{z}} = |\vec{\mathbf{A}}||\hat{\boldsymbol{z}}|\cos\varphi$ Using the fact that $|\hat{\boldsymbol{z}}| = \mathbf{1}$

$$\vec{\mathbf{a}} \bullet \hat{\boldsymbol{z}} = a_z = |\vec{\mathbf{a}}|\boldsymbol{cos}(\boldsymbol{\gamma}) \quad \textbf{z scalar component of vector } \vec{\mathbf{a}}$$

- Where the angle gamma (γ) is measured from the unit vector $\hat{\mathbf{z}}$ to the vector $\vec{\mathbf{a}}$. This is measured from the positive z-axis to the vector $\vec{a}$.



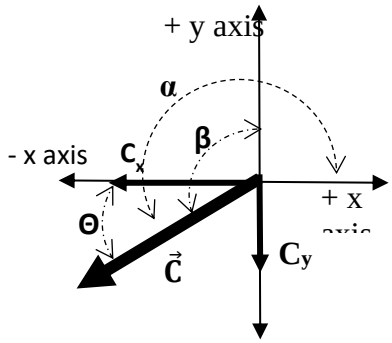
The components are determined by

$$a_x = |\vec{\mathbf{a}}|\cos(\alpha)$$
$$a_y = |\vec{\mathbf{a}}|\cos(\beta)$$
$$a_z = |\vec{\mathbf{a}}|\cos(\gamma)$$

α (alpha) is the angle measured from the positive *x*-axis to the vector ($\vec{\mathbf{a}}$)
β (beta) is the angle measured from the positive *y*-axis to the vector ($\vec{\mathbf{a}}$)
γ (gamma) is the angle measured from the positive *z*-axis to the vector ($\vec{\mathbf{a}}$)

Note: For the cosine function the direction the angle is measured (CW or CCW) does not affect the value of the scalar component of the vector. The acute angles are shown but the obtuse angle can also be used, the "cosine" value is the same.

Example: If displacement vector $\vec{\mathbf{C}}$ has a magnitude of 12 m and a direction Θ = 20° below the negative x-axis, what is vector $\vec{\mathbf{C}}$ in component form?



The angle to calculate the x scalar component of the vector is α and is measured from the positive x-axis to the vector,

therefore α = 180° + Θ = 200°

$$\vec{\mathbf{C}} \bullet \hat{\boldsymbol{x}} = C_x = |\vec{\mathbf{C}}|\cos(\alpha) = (12\text{m})\cos(200°) = -11.3 \text{ m}$$

The angle to calculate the y scalar component of the vector is β and is measured from the positive y-axis to the vector,

therefore β = 90° + Θ, β = 20° + 90° = 110°

$$C_y = |\vec{\mathbf{C}}|\cos(\beta) = (12\text{m})\cos(110°) = -4.1 \text{ m}$$

Vector $\vec{\mathbf{C}}$ in component form

$$\vec{\mathbf{C}} = (-11.3 \text{ m})\hat{\mathbf{x}} + (-4.1 \text{ m})\hat{\mathbf{y}}$$

Magnitude of a vector in vector component form is given by the Dot Product with itself, where the angle between two vectors that are equal is φ = 0° making cos(φ) = 1

$$\vec{\mathbf{A}} \bullet \vec{\mathbf{A}} = |\vec{\mathbf{A}}||\vec{\mathbf{A}}|\boldsymbol{cos}(\boldsymbol{\varphi}) \qquad \text{then} \qquad \vec{\mathbf{A}} \bullet \vec{\mathbf{A}} = |\vec{\mathbf{A}}|^2$$

Magnitude of a vector in component form (for unit vectors that are perpendicular)

$$\left|\vec{A}\right| = \sqrt{\vec{A} \bullet \vec{A}} = \sqrt{[(A_x)^2 + (A_Y)^2 + (A_z)^2]}$$

Direction of a vector in component form

$$\alpha = \cos^{-1}\left[\frac{A_x}{|\vec{A}|}\right] \qquad \beta = \cos^{-1}\left[\frac{A_y}{|\vec{A}|}\right] \qquad \gamma = \cos^{-1}\left[\frac{A_z}{|\vec{A}|}\right]$$

Example: Given the three-dimensional force vector below, what is the magnitude and direction of the vector?

$$\vec{F} = (40\text{ N})\hat{x} + (20\text{ N})\hat{y} + (-60\text{ N})\hat{z}$$

$$\left|\vec{F}\right| = \sqrt{\vec{F} \bullet \vec{F}} = \sqrt{\left[(F_x)^2 + \left(F_y\right)^2 + (F_z)^2\right]}$$

$F_x = 40$ N $F_y = 20$ N and $F_z = -60$N

$$\left|\vec{F}\right| = \sqrt{[(40\text{ N})^2 + (20\text{ N})^2 + (-60\text{N})^2]} = 74.8\text{ N}$$

Now the angles can be calculated using the magnitude of the vector and its components.

$$\alpha = \cos^{-1}\left[\frac{F_x}{\left|\vec{F}\right|}\right] = \cos^{-1}\left[\frac{40\text{N}}{74.8\text{N}}\right] = 57.7°$$

$$\beta = \cos^{-1}\left[\frac{F_y}{|\vec{F}|}\right] = \cos^{-1}\left[\frac{20\text{ N}}{74.8\text{N}}\right] = 74.5°$$

$$\gamma = \cos^{-1}\left[\frac{F_z}{|\vec{F}|}\right] = \cos^{-1}\left[\frac{-60\text{N}}{74.8\text{ N}}\right] = 143.3°$$

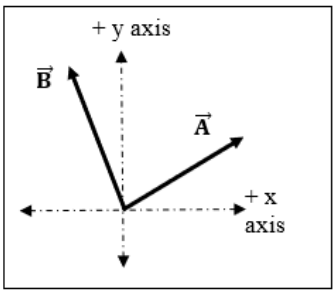
Note: Don't forget to include the sign (+ or -) when taking the inverse cosine.

Adding vectors Mathematically - When adding vector mathematically, the vector must be in component form. The components along each axis are added together creating the sum of the vector components along that axis. Each of the new components along with their axis unit vector form a new vector. This new vector is the sum of all vectors added and is the resultant vector in component form. The magnitude and direction of the resultant vector can be determined by the usual means of converting a vector from component form to magnitude and direction.

Mathematical Method for addition/subtraction Vectors in Components Form

For adding two vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ to obtain the resultant vector $\vec{\mathbf{R}}$

$$\vec{\mathbf{R}} = \vec{\mathbf{A}} + \vec{\mathbf{B}}$$



When adding vectors, all vectors must be in component form then the components of the vectors must be added to obtain the resultant vector.

All x-components of the vectors are added to get the x-component of the resultant vector ΣR_x.

All y-components of the vectors are added to get the y-component of the resultant vector. ΣR_y

$$\vec{\mathbf{R}} = \vec{\mathbf{A}} + \vec{\mathbf{B}}$$

$$\vec{\mathbf{R}} = (A_x + B_x + \cdots)\hat{\mathbf{x}} + (A_y + B_y + \cdots)\hat{\mathbf{y}}$$

Where

$\Sigma R_x = R_x$ = sum of all x-components = $A_x + B_x + \ldots$ for the above example
$\Sigma R_y = R_y$ = sum of all y-components = $A_y + B_y + \ldots$ for the above example

$$\vec{\mathbf{R}} = (\Sigma R_x)\hat{\mathbf{x}} + (\Sigma R_y)\hat{\mathbf{y}}$$

The scalar components of the vector along each axis must be added, this sum along each axis give the scalar components of the resultant vector.

Using the rules for taking the Dot Product of unit vectors

$\hat{\boldsymbol{x}}\cdot\hat{\boldsymbol{x}} = 1$	$\hat{\boldsymbol{x}}\cdot\hat{\boldsymbol{y}} = 0$
$\hat{\boldsymbol{y}}\cdot\hat{\boldsymbol{y}} = 1$	$\hat{\boldsymbol{x}}\cdot\hat{\boldsymbol{z}} = 0$
$\hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{z}} = 1$	$\hat{\boldsymbol{z}}\cdot\hat{\boldsymbol{y}} = 0$

- The projection of any vector along the unit vector direction can be determined using the Dot Product of the vector with the unit vector along that axis.

All scalar components of the protection of the vector are added to get the scalar x-component of the resultant vector.

$$\Sigma R_x = A_x + B_x + C_x + \ldots = \vec{\mathbf{R}}\cdot\hat{\mathbf{x}}$$

All scalar components of the protection of the vector are added to get the scalar y-component of the resultant vector.

$$\Sigma R_y = A_y + B_y + C_z + \ldots = \vec{\mathbf{R}} \bullet \hat{\mathbf{y}}$$

All scalar components of the protection of the vector are added to get the scalar z-component of the resultant vector.

$$\Sigma R_z = A_z + B_z + C_z + \ldots = \vec{\mathbf{R}} \bullet \hat{\mathbf{z}}$$

Where

$$A_x = \vec{A} \bullet \hat{x} = \left|\vec{A}\right| cos(\alpha_A) \qquad B_x = \vec{B} \bullet \hat{x} = \left|\vec{B}\right| cos(\alpha_B) \qquad C_x = \vec{C} \bullet \hat{x} = \left|\vec{C}\right| cos(\alpha_C)$$
$$A_y = \vec{A} \bullet \hat{y} = \left|\vec{A}\right| cos(\beta_A) \qquad B_y = \vec{B} \bullet \hat{y} = \left|\vec{B}\right| cos(\beta_B) \qquad C_y = \vec{C} \bullet \hat{y} = \left|\vec{C}\right| cos(\beta_C)$$
$$A_z = \vec{A} \bullet \hat{z} = \left|\vec{A}\right| cos(\gamma_A) \qquad B_z = \vec{B} \bullet \hat{z} = \left|\vec{B}\right| cos(\gamma_B) \qquad C_z = \vec{C} \bullet \hat{z} = \left|\vec{C}\right| cos(\gamma_C)$$

The resulting vector will have components that are the sum of all the vector's components along each axis.

$$\vec{\mathbf{R}} = (\Sigma R_x)\hat{\mathbf{x}} + \left(\Sigma R_y\right)\hat{\mathbf{y}} + (\Sigma R_z)\hat{\mathbf{z}}$$

The magnitude and direction of the resulting vector can be determined by

$$\vec{\mathbf{R}} = \sqrt{\vec{\mathbf{R}} \bullet \vec{\mathbf{R}}} = \sqrt{[(\Sigma R_x)^2 + (\Sigma R_Y)^2 + (\Sigma R_z)^2]}$$

$$\alpha = \cos^{-1}\left[\frac{\Sigma R_x}{|\vec{\mathbf{R}}|}\right] \qquad \beta = \cos^{-1}\left[\frac{\Sigma R_y}{|\vec{\mathbf{R}}|}\right] = \qquad \gamma = \cos^{-1}\left[\frac{\Sigma R_z}{|\vec{\mathbf{R}}|}\right]$$

Steps for Adding Vectors in component form

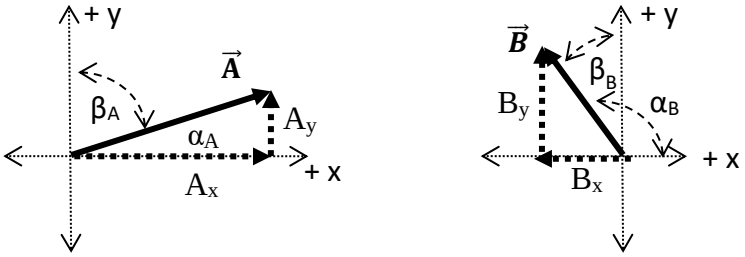
1) Determine what vectors are being summed and write the summation equation of all vectors.

$$\Sigma\vec{\mathbf{F}} = \vec{\mathbf{A}} + \vec{\mathbf{B}}$$

2) Calculate the components of the vectors given their magnitude and direction, Example below are the components for vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ along the x and y axis.

$$\vec{\mathbf{A}} = (A_x)\hat{x} + (A_y)\hat{y} \qquad \vec{\mathbf{B}} = (B_x)\hat{x} + (B_y)\hat{y}$$

$$\vec{\mathbf{A}} \bullet \hat{x} = A_x = \left|\vec{A}\right|\cos(\alpha_A) \qquad \vec{\mathbf{B}} \bullet \hat{x} = B_x = \left|\vec{B}\right|\cos(\alpha_B)$$
$$\vec{\mathbf{A}} \bullet \hat{y} = A_y = \left|\vec{A}\right|\cos(\beta_A) \qquad \vec{\mathbf{B}} \bullet \hat{y} = B_y = \left|\vec{B}\right|\cos(\beta_B)$$



- ❖ The angle alpha (α) is always measured from the positive x-axis to the vector.
- ❖ The angle beta (β) is always measured from the positive y-axis to the vector.
- ❖ The components are drawn from the base of the vector along the axis to the point where the projection of the tip of the vector would be located.
- ❖ Positive components point along the positive direction of the axis.
- ❖ Negative components point along the negative direction of the axis.

3) Write out vectors in component form from the components calculated.

$$\vec{\mathbf{A}} = (A_x)\hat{\boldsymbol{x}} + (A_y)\hat{\boldsymbol{y}} \qquad \vec{\mathbf{B}} = (B_x)\hat{\boldsymbol{x}} + (B_y)\hat{\boldsymbol{y}}$$

The resultant vector in vector component form is given as the sum of the components along the x and sum of the components along the y axis time the unit vector for that axis direction.

$$\boldsymbol{\Sigma\vec{F}} = (\Sigma F_x)\hat{\mathbf{x}} \; + \; (\Sigma F_y)\hat{\mathbf{y}}$$

ΣF_x is the sum of the forces along the x-axis
ΣF_y is the sum of the forces along the y-axis

4) Determine the components of the net force by summing the x-components of all vectors that are being added.

$$\Sigma\vec{\mathbf{F}} \bullet \hat{\mathbf{x}} = \Sigma F_x = A_x + B_x + \ldots$$

5) Determine the components of the net force by summing the y-components of all vectors that are being added.

$$\Sigma\vec{\mathbf{F}} \bullet \hat{\mathbf{y}} = \Sigma F_y = A_y + B_y + \ldots$$

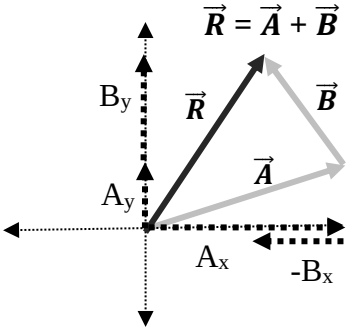
6) Put the net force vector in component form using the values of the x and y components.

$$\Sigma\vec{\mathbf{F}} = \vec{\mathbf{A}} + \vec{\mathbf{B}} = (\Sigma F_x)\hat{\mathbf{x}} \; + \; (\Sigma F_y)\hat{\mathbf{y}}$$

$$\Sigma\vec{\mathbf{F}} = (\Sigma F_x)\hat{\mathbf{x}} \; + \; (\Sigma F_y)\hat{\mathbf{y}}$$

7) Determining the Resultant component vector and its Magnitude and Direction.

$$\Sigma\vec{\mathbf{F}} = (\Sigma F_x)\hat{\mathbf{x}} \; + \; (\Sigma F_y)\hat{\mathbf{y}}$$



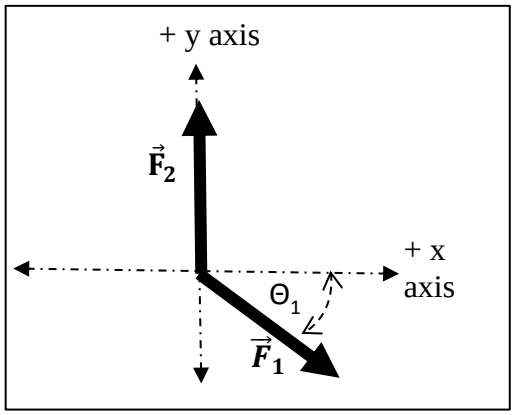
Magnitude $\left|\Sigma\vec{\mathbf{F}}\right| = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$

Direction

$$\alpha = \cos^{-1}\left[\frac{\Sigma F_x}{|\Sigma\vec{F}|}\right] \quad \beta = \cos^{-1}\left[\frac{\Sigma F_y}{|\Sigma\vec{F}|}\right]$$

❖ The components of any vector can be used in place of the vector itself in any calculation where it is convenient to do so!

Example: Two force vectors are acting on a box, the magnitude and directions of the vectors are given below, what is the magnitude of the net force acting on the box and in what direction (angles α and β)?



$\left|\vec{\mathbf{F}}_1\right| = 2.15\ N \quad \Theta_1 = 26.6°$

$\left|\vec{\mathbf{F}}_2\right| = 7.2\ N$

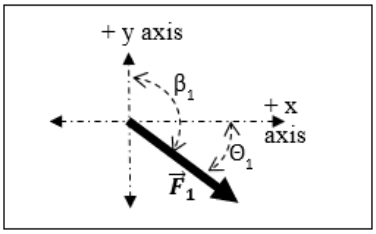
Along + y axis

$$\Sigma\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 \; = \; \text{net force}$$

1) Determine the components of the vector $\vec{\mathbf{F}}_1$ and put in component vector form.

Determine the angles α and β for force vector 1. Looking at vector 1 only

$\alpha_1 = \Theta_1 = 26.6°$ from positive x-axis
$\beta_1 = 90° + 26.6°$ from positive y-axis



$\vec{\mathbf{F}}_1 \bullet \hat{\mathbf{x}} = F_{1x} = \left|\vec{\mathbf{F}}_1\right|\cos(\alpha_1)$ $F_{1x} = \left|\vec{\mathbf{F}}_1\right|\cos(26.6°) =$ + 1.92 N

$\vec{\mathbf{F}}_1 \bullet \hat{\mathbf{y}} = F_{1y} = \left|\vec{\mathbf{F}}_1\right|\cos(\beta_1)$ $F_{1y} = \left|\vec{\mathbf{F}}_1\right|\cos(116.6°) =$ - 0.96 N

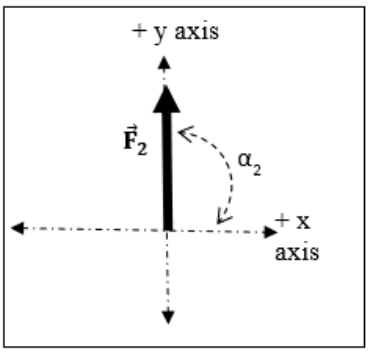
$\vec{\mathbf{F}}_1$ in component form is give as

$$\vec{F}_1 = (1.92\ \text{N})\hat{x} + (-\ 0.96\ \text{N})\hat{y}$$

2) Determine the components of the vector $\vec{\mathbf{F}}_2$ and put in component vector form.

Determine the angles α and β for force vector 2. Looking at vector 2 only

$\alpha_2 = 90°$ from positive x-axis
$\beta_2 = 0°$ along positive y-axis



$\vec{\mathbf{F}}_2 \bullet \hat{\mathbf{x}} = F_{2x} = |\vec{\mathbf{F}}_2|\cos(\alpha_2)$ $F_{2x} = |\vec{\mathbf{F}}_2|\cos(90°) = 0$ N

$\vec{\mathbf{F}}_2 \bullet \hat{\mathbf{y}} = F_{2y} = |\vec{\mathbf{F}}_2|\cos(\beta_2)$ $F_{2y} = |\vec{\mathbf{F}}_2|\cos(0°) =$ + 7.2 N

$\vec{\mathbf{F}}_2$ in component form is give as

$$\vec{F}_2 = (0\text{ N})\hat{x} + (7.2\text{ N})\hat{y}$$

3) Sum the forces acting along the x-axis.

$\Sigma F_x = F_{1x} + F_{2x} =$ + 1.92 N + 0 N = + 1.92 N

4) Sum the forces acting along the y-axis.

$\Sigma F_y = F_{1y} + F_{2y} =$ (- 0.96 N) + 7.2 N = + 6.24 N

5) Put the net force vector in component form.

$$\boldsymbol{\Sigma\vec{F}} = (\Sigma F_x)\hat{\mathbf{x}} + (\Sigma F_y)\hat{\mathbf{y}}$$

$$\boldsymbol{\Sigma\vec{F}} = (1.92\text{ N})\hat{\mathbf{x}} + (6.24\ N)\hat{\mathbf{y}}$$

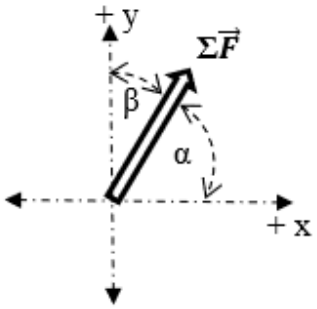
6) Determine the magnitude of the net force is given by

$$|\Sigma\vec{F}| = \sqrt{\Sigma F_x^2 + \Sigma F_y^2} = 6.5\ N$$

7) Determine the direction of the net force is given by

$$\alpha_{net} = cos^{-1}\left(\frac{\Sigma F_x}{|\Sigma\vec{F}|}\right) = 73° \qquad \beta_{net} = cos^{-1}\left(\frac{\Sigma F_y}{|\Sigma\vec{F}|}\right) = 17°$$

Therefore, the box experiences a force with a magnitude of 6.5 N at an angle of 73° measured from the positive x-axis due the two force vectors acting on it.



<u>Calculus and vectors in Component form</u>: For a vector that has a time (t) or displacement (x) dependance, the derivative or integral is taken on the scalar component along the $\hat{x}$, $\hat{y}$ and $\hat{z}$ directions independently for each scalar component of the vector in the Cartesian coordinate system.

Below is vector $\vec{A}$ in component form.

$$\vec{A} = (A_x)\hat{x} + (A_y)\hat{y} + (A_z)\hat{z}$$

The derivative of vector $\vec{A}$ with respect to time (or any other quantity) is the derivative of each of the scalar components of the vector.

$$\frac{d\vec{A}}{dt} = \left(\frac{dA_x}{dt}\right)\hat{x} + \left(\frac{dA_y}{dt}\right)\hat{y} + \left(\frac{dA_z}{dt}\right)\hat{z}$$

The integral of vector $\vec{A}$ with respect to time (or any other quantity) is the integral of each of the scalar components of the vector.

$$\int \vec{A}dt = \left(\int A_x dt\right)\hat{x} + \left(\int A_y dt\right)\hat{y} + \left(\int A_z dt\right)\hat{z}$$

Example: Given the velocity vector of a drone with a time dependance below.

$$\vec{v} = ([0.3\ m/s^3]\,t^2)\hat{x} + (5\ m/s + [4\,m/s^2]t)\hat{y}$$

A) Determine the magnitude and directions of the displacement of drone from 0s to 10s.

$$\int \vec{v}dt = \left(\int v_x dt\right)\hat{x} + \left(\int v_y dt\right)\hat{y}$$

$$\vec{x} = \int_{0s}^{10s} \vec{v}dt = \int_{0s}^{10s} \{([0.3\ m/s^3]\,t^2)\hat{x} + (5\ m/s + [4.0\,m/s^2]t)\hat{y}\}dt$$

$$\vec{x} = \int_{0s}^{10s} \vec{v}dt = \left\{\left(\int_{0s}^{10s} 0.3\ m/s^3]\, t^2 dt\right)\hat{x} + \left(\int_{0s}^{10s} 5\ m/s\, dt + \int_{0s}^{10s} [4.0\, m/s^2]t\right)\hat{y}\right\}$$

$$\vec{x} = \int_{0s}^{10s} \vec{v}dt = \left\{\left(\left[\frac{0.3\, m/s^3\ t^3}{3}\right]_{0s}^{10s}\right)\hat{x} + \left([(5\ m/s)t]_{0s}^{10s} + \left[\frac{4.0\, m/s^2\ t^2}{2}\right]_{0s}^{10s} t\right)\hat{y}\right\}$$

$$\vec{x} = (100\ m)\hat{x} + (250\ m)\hat{y}$$

$$|\vec{x}| = 269\ m \qquad \alpha = 68° \qquad \beta = 22°$$

B) Determine the magnitude of the acceleration of the drone at 10s.

$$\vec{a} = \frac{d\vec{v}}{dt} = \left(\frac{dv_x}{dt}\right)\hat{x} + \left(\frac{dv_y}{dt}\right)\hat{y} = (0.6\ m/s^3 t)\hat{x} + (4\ m/s^2)\hat{y}$$

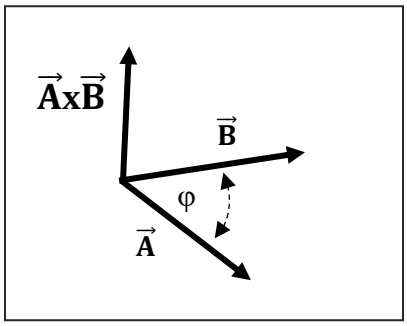
$$\vec{a} = (6\ m/s^2)\hat{x} + (4\ m/s^2)\hat{y}$$

$$|\vec{a}| = 7.2\ m/s^2$$

<u>Vector Product or Vector Cross Product</u> (**"x"**)– the multiplication of the vectors that produces a third vector quantity that is perpendicular to the other two

The new vector $\vec{\mathbf{C}}$ is given by

$$\vec{\mathbf{C}} = \vec{\mathbf{A}}\mathbf{x}\vec{\mathbf{B}} = (|\vec{\mathbf{A}}||\vec{\mathbf{B}}|\sin\varphi)\hat{\mathbf{C}}$$



$|\vec{\mathbf{A}}|$ – is the magnitude of vector $\vec{\mathbf{A}}$
$|\vec{\mathbf{B}}|$ – is the magnitude of vector $\vec{\mathbf{B}}$
φ – is the angle between vectors $\vec{\mathbf{A}}$ **and** $\vec{\mathbf{B}}$
$\hat{\mathbf{C}}$ **-** is the unit vector in the direction of the vector $\vec{\mathbf{C}}$ and is perpendicular to both vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$

The magnitude of vector $\vec{\mathbf{C}}$ is given by $|\vec{\mathbf{C}}| = |\vec{\mathbf{A}}||\vec{\mathbf{B}}|\sin\varphi$
The direction of vector is $\vec{\mathbf{C}}$ is given by the unit vector $\hat{\mathbf{C}}$

- Pronounced "A cross B" or "the cross product of A and B"
- The Cross Product produces a vector quantity and therefore must be treated as a new vector with both magnitude and direction.
- **Cross Products create vectors, Dot Products create scalars**

❖ The angle of the new vector ($\vec{\mathbf{C}}$) is 90° to relative to both vector $\vec{\mathbf{A}}$ and vector $\vec{\mathbf{B}}$.

The angle phi (φ) and the Cross Product

When φ = 0° or 360° then $|\vec{C}| = \vec{A}x\vec{B} = 0$

When φ = 90° or 270° then $|\vec{C}| = \vec{A}x\vec{B} = |A||B|$ = maximum

Direction of the Cross Product (Right Hand Rule):

If $\vec{c} = \vec{a}x\vec{b}$

1) Point your index fingers long the first vector in the cross product ($\vec{a}$)
2) Point your next three fingers along the second vector in the cross product ($\vec{b}$)
3) your thumb point in the direction of the third vector that is the cross product ($\vec{c}$)

